Performing a Quantum Subroutine Using Cluster States

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Cluster States

- A cluster state is a type of highly entangled state of multiple qubits [1]. It reduces operational overhead.
- Optical cluster states are a proposed tool to achieve quantum computational universality in linear optical quantum computing (LOQC) [2].
- Other hardware for generating cluster states is Superconducting qubits [3], Trapped Ion [4], Neutral Atoms [5].
- For measurement based quantum computing (MBQC) photonic systems are mostly promising [6].
- **To mention that** we will not be discussing about how to generate cluster state with different hardware.

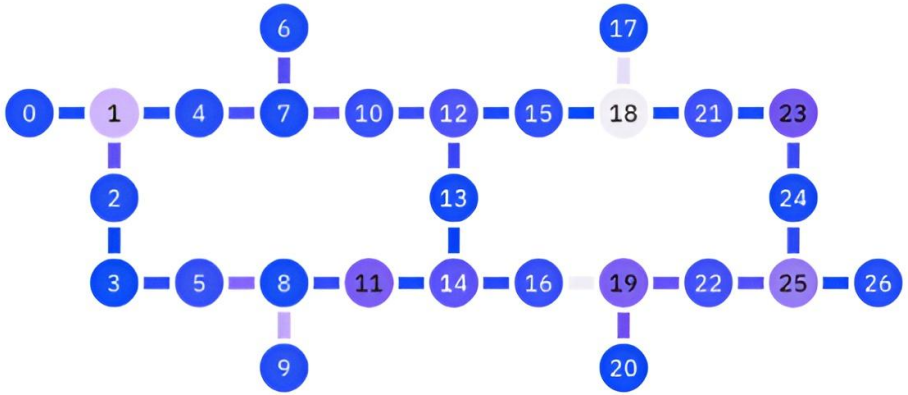


Figure 1: The qubits are entangled in a specific configuration to form a cluster state. The image is sourced from the IBM Quantum Processing Unit (QPU) documentation, specifically pertaining to the ibm_brisbane hardware architecture.

Quantum Subroutine

- Subroutines are the basic building blocks of computer programs.
- As classical algorithm got subroutine so as quantum algorithm.
- Quantum subroutine modular block of a quantum algorithm, reusable like classical functions.
- Examples: Quantum Fourier Transform (QFT), Grover's Oracle, Entanglement modules.



Figure 2: Quantum subroutines, like LEGO blocks, are fundamental building units of quantum algorithms. Tuxedo cat image sourced from Amazon. .

Research Problem

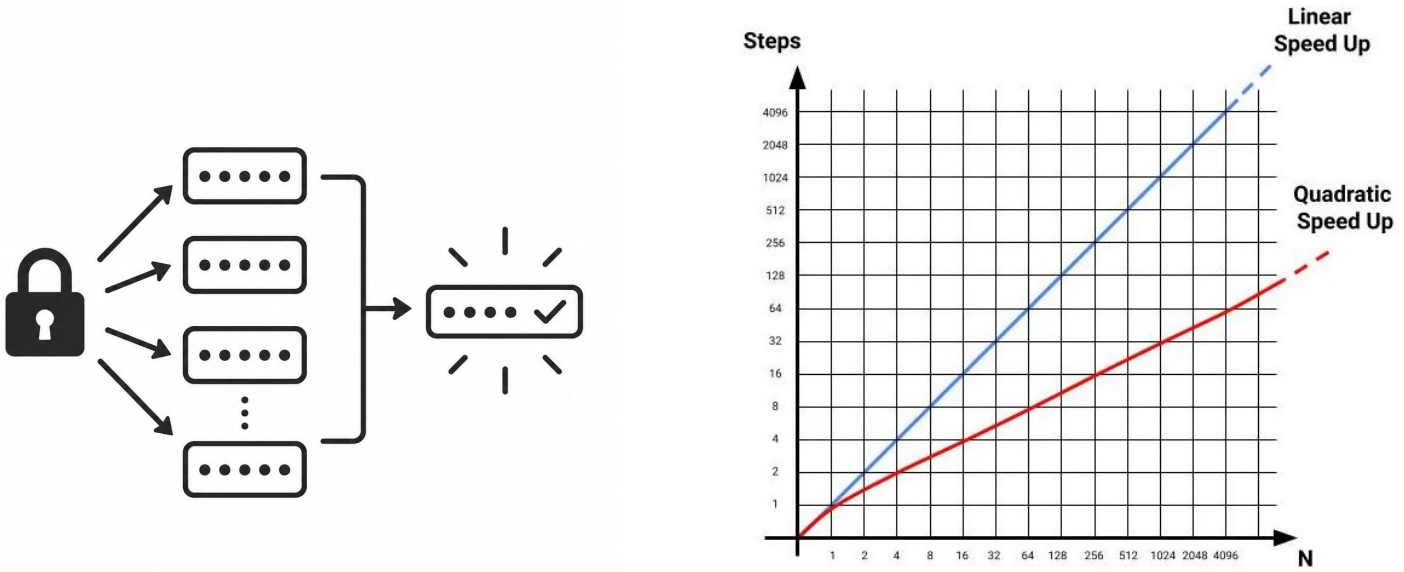
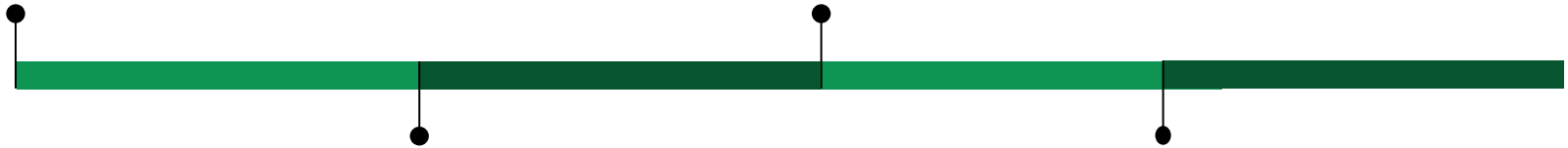


Figure 3**: (Left)** Illustration of a classical unstructured search problem, where the goal is to find the correct password to unlock a mobile phone. **(Right)** For unstructured search, Grover's algorithm provides a quadratic speedup over classical brute-force search.

Motivation and Objectives

Creating 2D cluster state: Generation of a one and two dimensional cluster state

Creating Grover's oracle for different input: Designing Grover's oracle tailored to different search inputs.



Creating gate like functionality: By selecting appropriate measurement bases on the cluster state.

Applying oracle to the Grover's algorithm: Integrating the constructed oracle into Grover's search algorithm for execution

Background: Quantum Computation Models

- A **qubit** is a two-level quantum system, meaning it can exist in a superposition of two basis states, typically labeled $|0\rangle$ and $|1\rangle$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

as α and β are complex numbers with two degrees of freedom each which can be written as,

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

The parameters θ and ϕ, re-interpreted in spherical coordinates as respectively the colatitude with respect to the z-axis and the longitude with respect to the x-axis

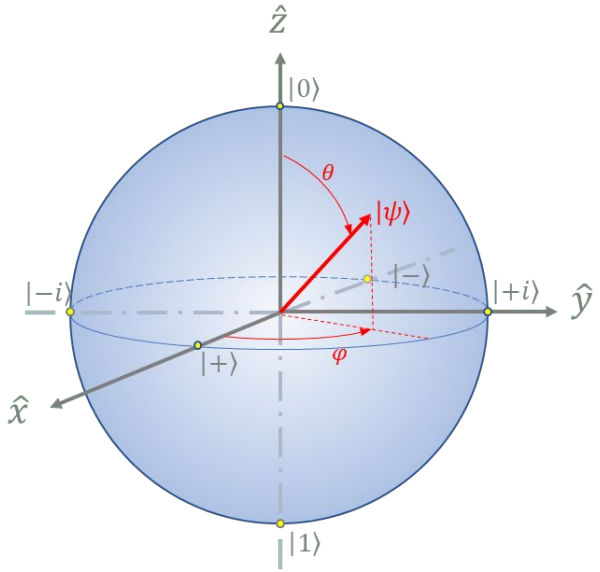
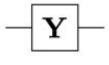
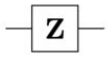
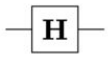
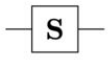
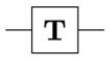
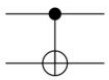


Figure 4: The Bloch sphere representation utilizes the parameters θ and ϕ, which can be reinterpreted in spherical coordinates as the colatitude (θ) measured from the z-axis and the longitude (ϕ) measured from the x-axis. Bloch sphere image is taken from wikipedia.

Background: Quantum Computation Models

Table 1: Various quantum logic gates (Table is taken from wikipedia)

Operator	Gate(s)	Matrix
Pauli-X (X)		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)		$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

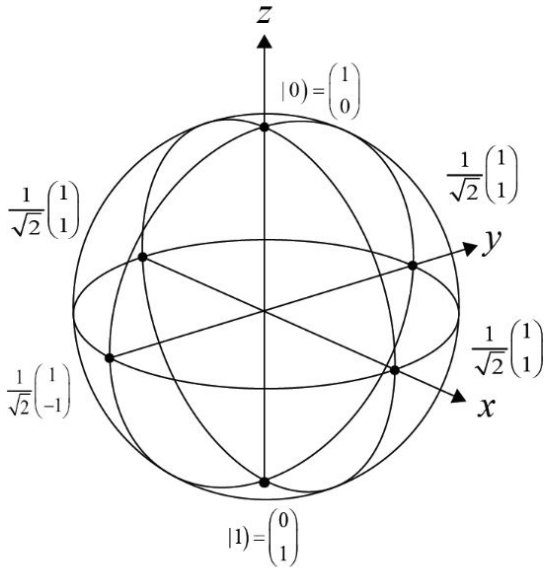


Figure 5: The Bloch sphere illustrates state rotations corresponding to the implementation of various quantum logic gates. Image is taken from the book "Artificial Intelligence By Example".

Working Principle of MBQC

- Measurement-based quantum computer (MBQC) also known as one-way quantum computer.
- It is a method of quantum computing that first prepares an entangled resource state, usually a cluster state or graph state, then performs single qubit measurements on it.
- It is "one-way" because the resource state is destroyed by the measurements.

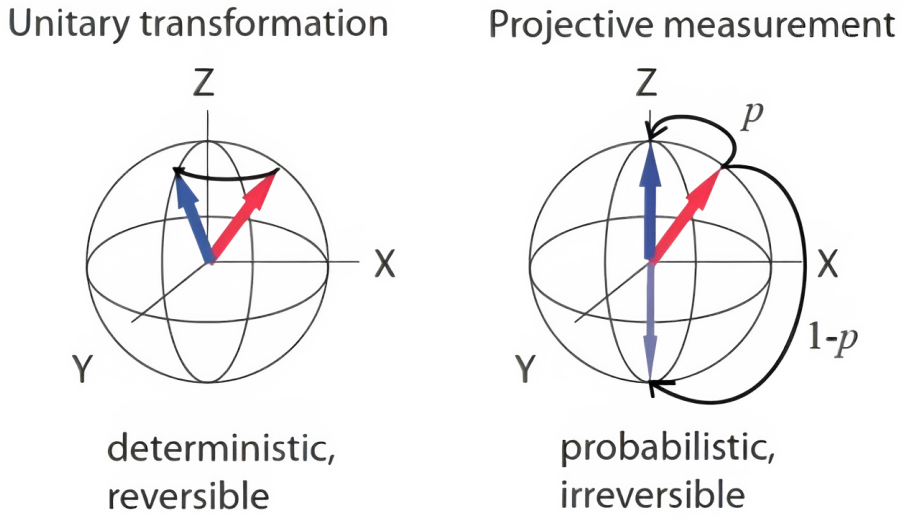


Figure 6: The two Bloch spheres highlight different aspects of gate-based and measurement-based models of quantum computation. Bloch sphere image is taken from Robert Raussendorf lecture series.

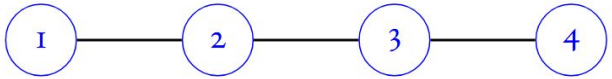
Mathematical Construction of Cluster States

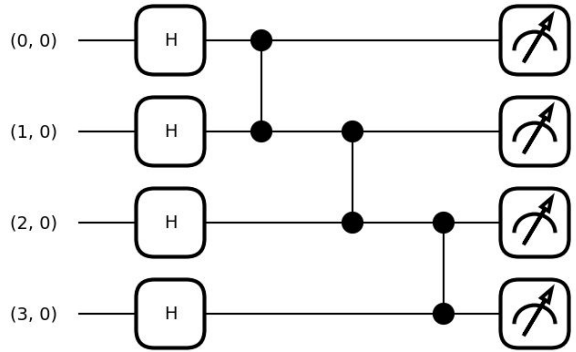
- In quantum information and quantum computing, a cluster state is a type of highly entangled state of multiple qubits. Formally, cluster states are states which obey the set eigenvalue equations,

$$K^{(a)}|\phi_{\{k\}}\rangle_C = (-1)^{ka}|\phi_{\{k\}}\rangle_C$$

Where, $K^{(a)}$ is the correlation operator and can be written as,

$$K^{(a)} = \sigma_x^{(a)} \bigotimes_{b \in N(a)} \sigma_z^{(b)}$$





$$|\phi\rangle_c = \frac{1}{2}(|+\rangle_1|0\rangle_2|+\rangle_3|0\rangle_4 + |+\rangle_1|0\rangle_2|-\rangle_3|1\rangle_4 + |-\rangle_1|1\rangle_2|-\rangle_3|0\rangle_4 + |-\rangle_1|1\rangle_2|+\rangle_3|1\rangle_4)$$

Figure 7: (Left) A 4-qubit cluster state along with its corresponding mathematical representation; (Right) the quantum circuit used to generate this cluster state. Circuit is generated using pennylane and cluster using latex

Information Propagation in MBQC

- Measurement-based quantum computation heavily relies on the idea of information propagation. In particular, we make use of a protocol called quantum teleportation.
- Alice has qubit $|\psi\rangle$ that she wants to teleport.
- Alice and Bob share an entangled Bell pair $|\Phi+\rangle$
- Alice measures her two qubits and sends classical bits to Bob.
- Bob applies a Pauli operation and get's the qubit $|\psi\rangle$

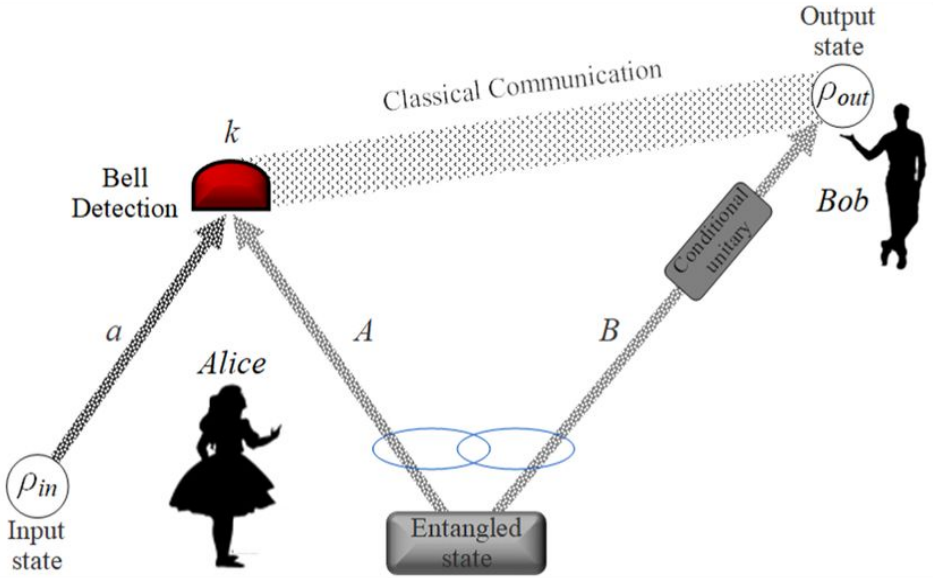


Figure 8: An illustration of quantum teleportation, where Alice and Bob share an entangled state to facilitate the transfer of quantum information [8].

Universality of MBQC

- To show that our measurement scheme is just as power as gate based counterpart. Proven statements are –
- How information propagates through the cluster state.
- How arbitrary single-qubit rotations can be implemented
- How a two-qubit gate can be implemented in this scheme.

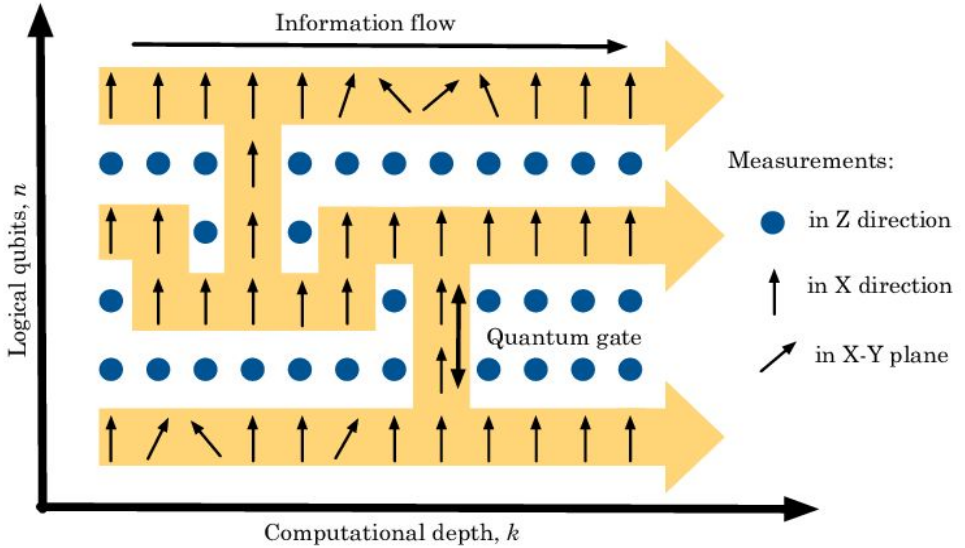


Figure 9: Measurement-based quantum computation (MBQC) illustrating the same principle through entanglement and adaptive measurements. Circuit and image is taken from pennylane documentation [1].

Single Qubit Rotations for MBQC

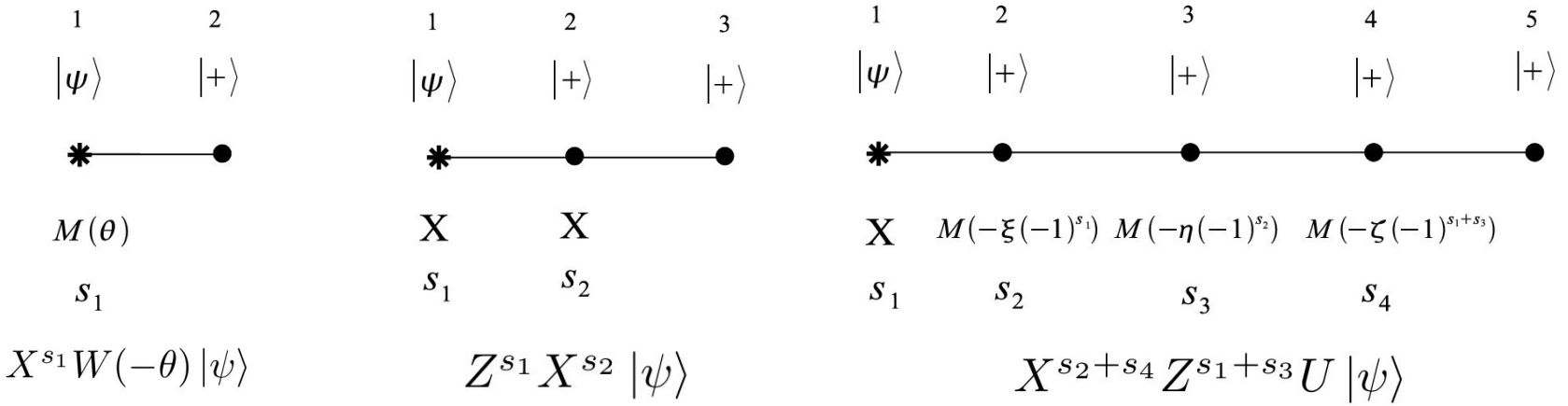


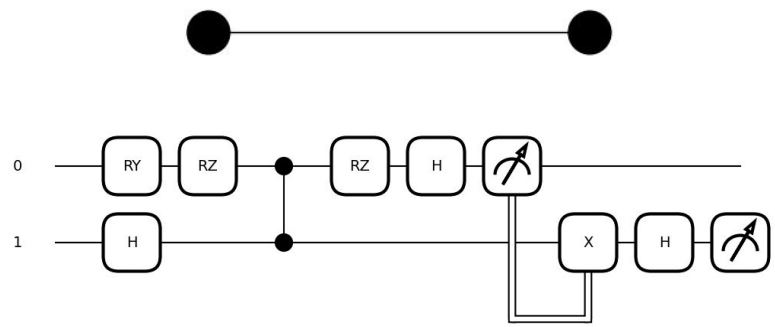
Figure 10: Represents how single qubit rotations works for MBQC. This illustrations are taken from "An introduction to measurement based quantum computation" by Richard Jozsa.

- Any quantum logic gate can be represented using a rotation gate, as shown here, where U denotes the unitary operation acting on the initial qubit.

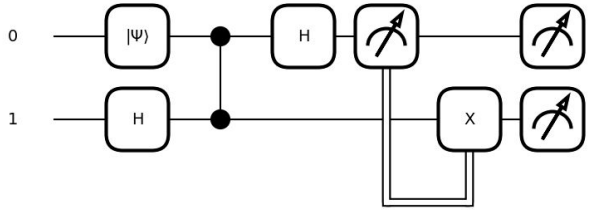
$$U = R_x(\zeta) R_z(\eta) R_x(\xi) \quad \text{for some } \xi, \eta, \zeta. \quad \xi,\ \eta, \text{ and } \zeta \text{ are Euler angles for } U$$

$$M(\theta) = \{|0\rangle \pm e^{i\theta}\,|1\rangle\}$$

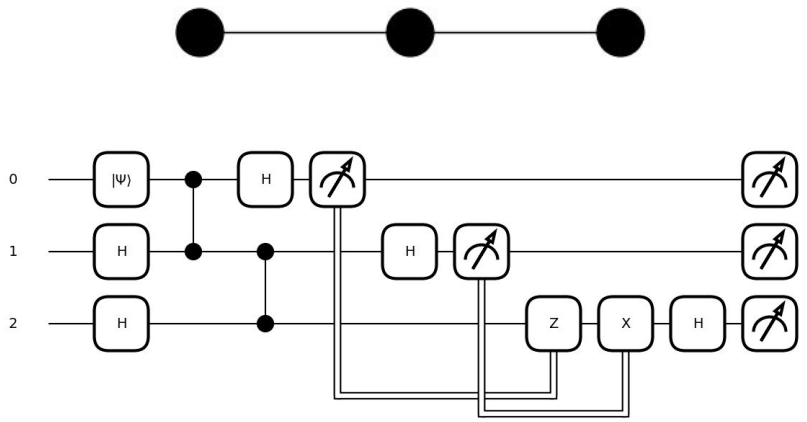
Single Qubit Rotation Circuit for MBQC Formalism



$$RZ(\theta)|\psi_{\text{in}}\rangle_1 = X^{s^0}\langle\pm|_0 RZ_0(\theta)CZ_{0,1}|+\rangle_1 \otimes |\psi_{\text{in}}\rangle_0$$



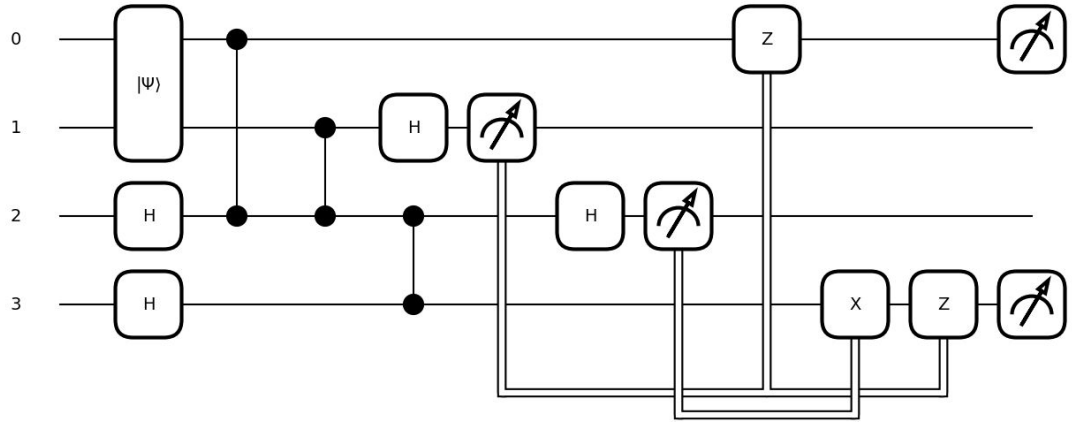
$$H|\psi_{\text{in}}\rangle_1 = X^{s^0}\langle\pm|_0 CZ_{0,1}|+\rangle_1 \otimes |\psi_{\text{in}}\rangle_0$$

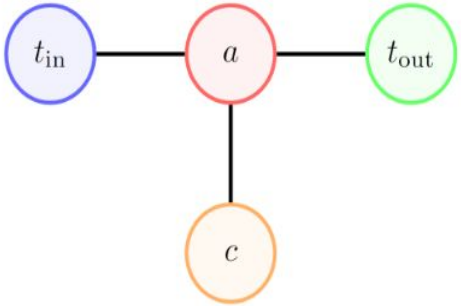


$$H|\psi_{\text{in}}\rangle_2 = Z_2^{s_0} X_2^{s_1} \langle\pm|_1\langle\pm|_0 CZ_{1,2}CZ_{0,1}|+\rangle_2|+\rangle_1|\psi\rangle_0$$

Figure 11: (left) Unitary operations for a 2-qubit cluster, including RZ and Hadamard gates (H), along with the corresponding circuit diagram. (right) Hadamard operation and circuit diagram for a 3-qubit cluster. Generated circuit and cluster with pennylane.

Two Qubit Gate Circuit for MBQC Formalism

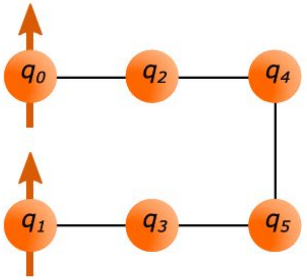




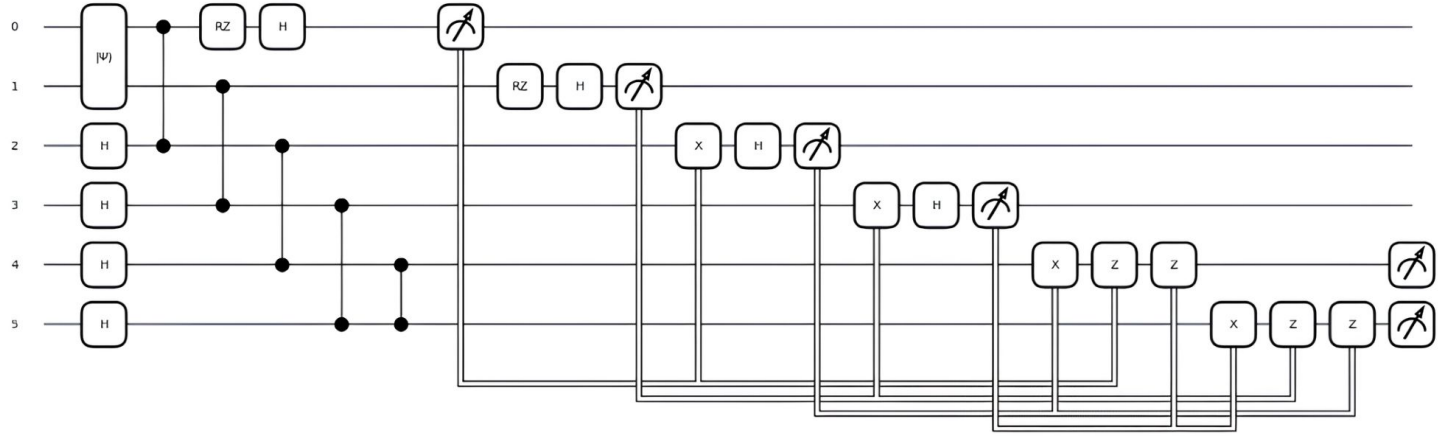
$$CNOT_{0,3}|\psi_{in}\rangle_{03} = X_3^{s_2} Z_3^{s_1} Z_0^{s_1} \langle \pm|_2 \langle \pm|_1 CZ_{0,2} CZ_{2,3} CZ_{1,2} |+\rangle_3 |+\rangle_2 |\psi_{in}\rangle_{01}$$

Figure 12: (left) CNOT operation along with its unitary operation equation; (right) corresponding cluster representation. Generated circuit and cluster with pennylane.

MBQC CZ Gate & Subroutine



Qubit	Stabilizer $k^{(q_i)}$
q_0	$\sigma_x^{(q_0)}\sigma_z^{(q_2)}$
q_1	$\sigma_x^{(q_1)}\sigma_z^{(q_3)}$
q_2	$\sigma_x^{(q_2)}\sigma_z^{(q_0)}\sigma_z^{(q_4)}$
q_3	$\sigma_x^{(q_3)}\sigma_z^{(q_1)}\sigma_z^{(q_5)}$
q_4	$\sigma_x^{(q_4)}\sigma_z^{(q_2)}\sigma_z^{(q_5)}$
q_5	$\sigma_x^{(q_5)}\sigma_z^{(q_3)}\sigma_z^{(q_4)}$



$$|\psi\rangle = \frac{1}{8}|000000\rangle + \frac{1}{8}|000001\rangle + \frac{1}{8}|000010\rangle + \frac{1}{8}|000011\rangle + \frac{1}{8}|000100\rangle - \frac{1}{8}|000101\rangle + \ldots$$
$$-\frac{1}{8}|111011\rangle - \frac{1}{8}|111100\rangle - \frac{1}{8}|111100\rangle + \frac{1}{8}|111101\rangle + \frac{1}{8}|111110\rangle - \frac{1}{8}|111111\rangle$$

Figure 13: (left) Six qubit cluster for CZ gate and it's correlation operators; (right) corresponding circuit representation for CZ MBQC and final quantum state for the six qubits. Generated using pennylane.

Methodology

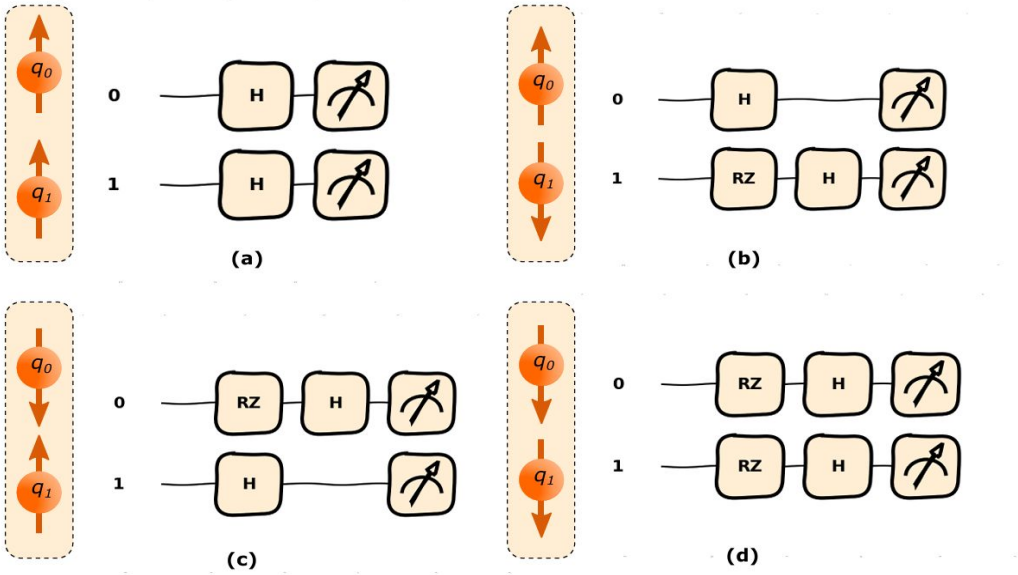


Figure 14: (left) Subroutine Construction based on the input qubit 0 and 1 with rotation defined on the XY plane (a) XY(0) at qubit 0 and 1 for quantum state $|11\rangle$ (b) XY(0) and XY(180) for input qubit 0 and 1 for quantum state $|10\rangle$ (c) XY(180) and XY(0) for input qubit 0 and 1 for quantum state $|10\rangle$ (d) XY(180) and XY(180) for input qubit 0 and 1 for quantum state $|00\rangle$

Simulation Result

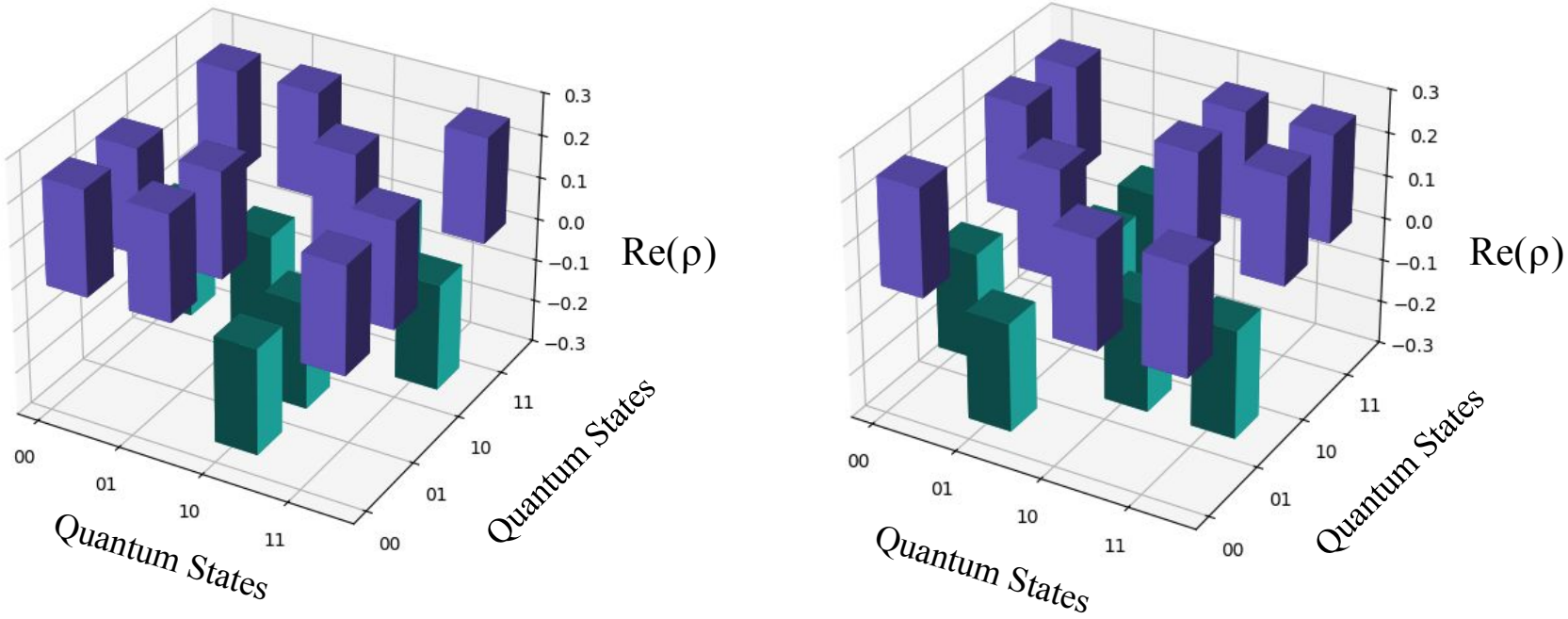


Figure 15: Density matrix verification for the initial state $|+\rangle \otimes |-\rangle$ where density matrix for (a) $|11\rangle$ (b) $|00\rangle$ and Re(ρ) is the real part of the density matrix.

Simulation Result

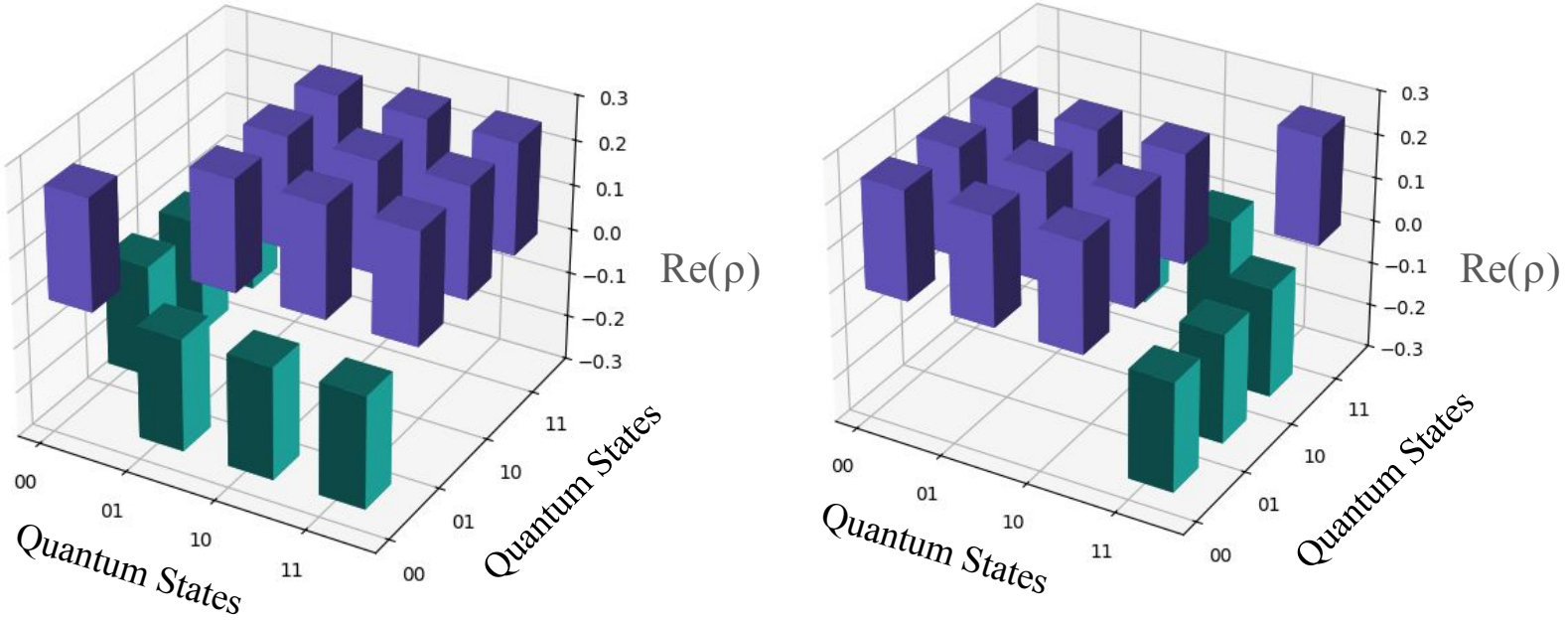


Figure 16: Density matrix verification for the initial state $|+\rangle \otimes |-\rangle$ where density matrix for (c) $|10\rangle$ (d) $|01\rangle$ and Re(ρ) is the real part of the density matrix.

Future Work

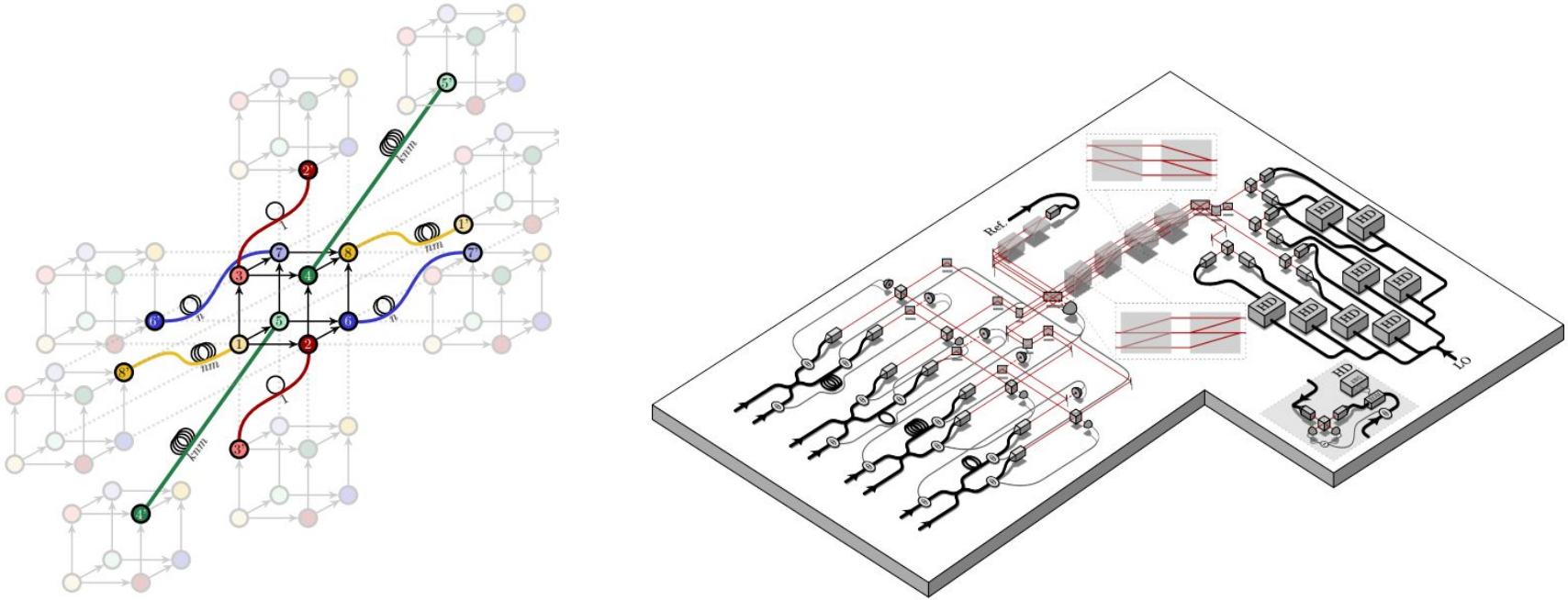


Figure 17: (Left) 3D cluster state; (Right) Photonic Integrated Circuit

Work Highlights




IEEE CS BDC SUMMER SYMPOSIUM 2025

Certificate of Contribution

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Figure 18: (Left) Won best poster award at ICEI 2026 (middle) IEEE CS summer symposium (Last) Under review on Quantum Information Processing (Springer Nature)

Acknowledgements





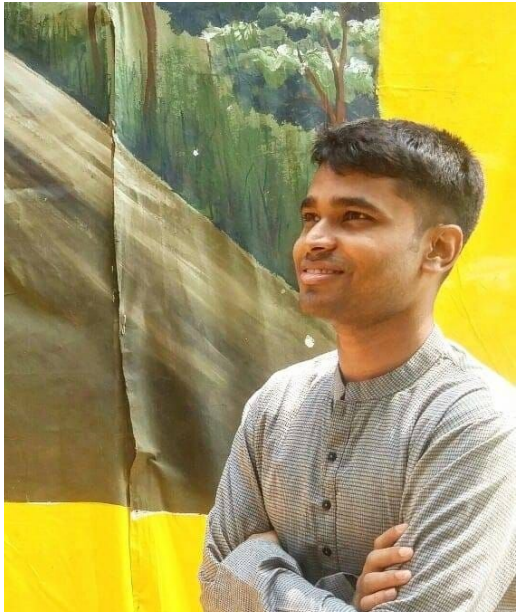


Figure 19: From left: Dr. Rasedujjaman Associate Professor, SUST, EEE, Dr. Mohsinur Rahman Adnan, Assistant Professor, BUET, EEE , and Sakibul Islam Sazzad (Ph.D. student at UCF).

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